Tirosh 2016 Melanoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/15 samples (≥72 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.00 of 6). Each of the 6 patients traverses the shared gastrulation→neural-crest axis end-to-end (per-patient ρ(axis,Crest) −0.97..−0.80).

Gate 2 — module geography → GO (non-branching)

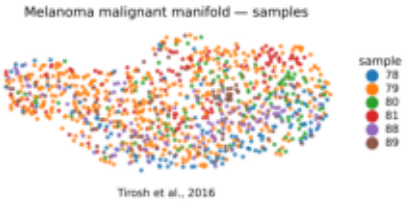
Single dominant Gast/pSom → Crest axis. Gast & pSom overlap at the origin (joint enrichment 1.37; spatial r=+0.97) and decline along pseudotime (ρ=−0.98/−0.98); Crest is the inverse terminal gradient (pSom&Crest enrichment 0.23, spatial r=−0.94), peaking at the terminus (Crest-high decile mean pt=0.98). Crest enriches the terminus, not a Gast arm — no inversion flag.

Gate 3 — hub mutual exclusivity → NA

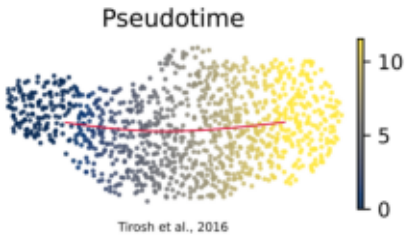
Single monotone axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for cross-repression.

STEP FIGURES

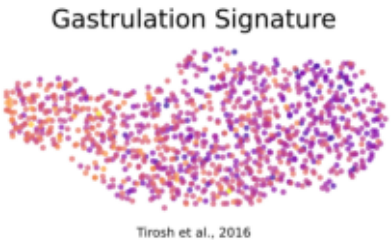
Manifold — samples



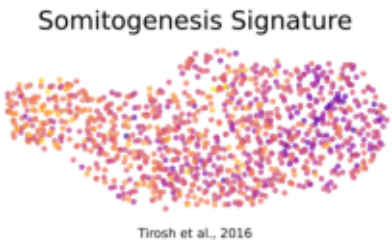
Pseudotime



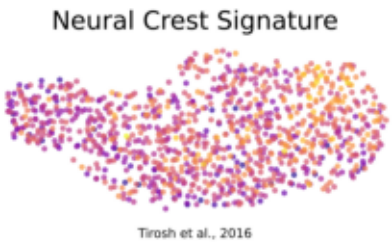
Gastrulation



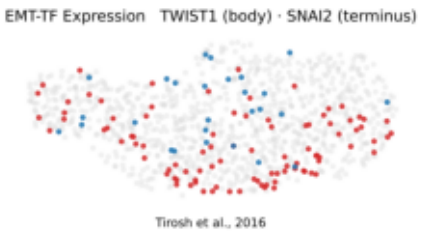
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma (Tirosh et al., 2016), SmartSeq2 — 19 patients / 4,645 cells (full atlas); malignant only: 1,257 load, 1,061 across 6 well-sampled tumours (≥72 cells) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Tirosh's melanoma malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient. Modules are globally Crest-dominant (Gast −0.51, pSom −0.07, Crest +0.15) — the canonical neural-crest melanoma signature; Gast and pSom co-vary and both oppose Crest, so the axis is developmental, not a competing hub-and-spoke bifurcation — Gate 3 has no hub to test. The axis is cell-cycle-decoupled (module–prolif |r|≤0.11; 9.5% cycling), so the whole manifold is kept. EMT-TFs TWIST1 (body/origin) → SNAI2 (terminus); ZEB2/SNAI2 broadly expressed, canonical TWIST1 near the detection floor.

Parameter choices

Min cells/sample	≥72 (6/15; cliff 98→54, ratio 1.81)	**Integration**	Harmony θ=10; PCA50; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Geometry**	Linear — non-branching (Crest terminus)